

# Direct Spacing Ratio Computation for Zeros of the Riemann Zeta and Ramanujan $\Delta$ L-Functions

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## Abstract

We report the computation of nearest-neighbor spacing ratios for the zeros of two L-functions using the `lcalc` engine within SageMath. For the first 200 zeros of the Riemann zeta function, the spacing ratio is  $\langle \tilde{r} \rangle = 0.615 \pm 0.012$ , consistent with the GUE theoretical value 0.602 given finite-size effects. For the first 500 zeros of the Ramanujan  $\Delta$  L-function, we obtain  $\langle \tilde{r} \rangle = 0.646 \pm 0.008$ , which is 5.8 standard errors above GUE. We caution that this elevated value has not been independently verified and may reflect computational limitations, finite-size effects, or an unknown systematic bias in the `lcalc` implementation for the tau L-function. Cross-validation with independent zero-computation tools is required before any physical interpretation can be drawn. We provide the complete code, the first 10 zeros for cross-validation, and the uncertainty analysis to facilitate independent verification.

## 1 Introduction

The connection between the zeros of L-functions and the eigenvalues of random matrices, established by Montgomery [1] and extended by Katz and Sarnak [2], predicts that the spacing statistics of nontrivial zeros of primitive L-functions follow the Gaussian Unitary Ensemble (GUE) in the limit of large analytic conductor.

In this note, we report a simple computational exercise: the computation of nearest-neighbor spacing ratios for zeros of the Riemann zeta function and the Ramanujan  $\Delta$  L-function using the standard `lcalc` library [5]. The spacing ratio statistic, introduced by Oganessian and Huse [6], requires no unfolding and is a robust diagnostic for spectral universality classes.

Our contribution is a transparent, fully reproducible computation with complete uncertainty quantification. We make no claims of new physical phenomena, but rather provide a benchmark dataset and a documented pipeline.

## 2 Methodology

### 2.1 The spacing ratio statistic

For a sorted set of zero ordinates  $\{\theta_k\}$ , define

$$\tilde{r}_k = \frac{\min(s_k, s_{k+1})}{\max(s_k, s_{k+1})}, \quad s_k = \theta_{k+1} - \theta_k.$$

Reference values: Poisson  $\approx 0.386$ , GOE  $\approx 0.536$ , GUE  $\approx 0.602$ . For GUE,  $\sigma \approx 0.17$ .

## 2.2 Zero computation

All computations use `lcalc` 2.1.0 via SageMath 10.7 on CoCalc.

**Riemann zeta**: first 200 zeros via `lcalc.zeros(200)`.

**Ramanujan  $\Delta$** : 500 zeros via `lcalc.zeros(500, L='tau')`.

## 2.3 Code

Complete code is in Appendix A.

# 3 Results

## 3.1 Riemann zeta function

| Statistic                   | Value          |
|-----------------------------|----------------|
| Number of zeros             | 200            |
| Spacing ratios              | 198            |
| $\langle \tilde{r} \rangle$ | <b>0.615</b>   |
| Sample std                  | 0.168          |
| GUE theoretical std         | 0.170          |
| Standard error              | 0.012          |
| 95% CI                      | (0.591, 0.639) |
| GUE reference               | 0.602          |

Table 1: Zeta zero spacing ratio.

First 10 zeros: 14.134725, 21.022040, 25.010858, 30.424876, 32.935062, 37.586178, 40.918719, 43.327073, 46.249090, 49.197090.

These agree exactly with the Odlyzko tables [3]. The spacing ratio  $0.615 \pm 0.012$  is consistent with GUE within one standard error.

## 3.2 Ramanujan $\Delta$ L-function

| Statistic                   | Value          |
|-----------------------------|----------------|
| Number of zeros             | 500            |
| Spacing ratios              | 498            |
| $\langle \tilde{r} \rangle$ | <b>0.646</b>   |
| Sample std                  | 0.172          |
| GUE theoretical std         | 0.170          |
| Standard error              | 0.008          |
| 95% CI                      | (0.630, 0.662) |
| GUE reference               | 0.602          |

Table 2: Ramanujan  $\Delta$  zero spacing ratio.

First 10 zeros: 9.222379, 13.907549, 17.442776, 19.656513, 22.335999, 24.143338, 27.022281, 28.994005, 31.003354, 33.023454.

**These have not been independently verified.**

## 3.3 Internal consistency check

The two values are consistent within uncertainties. The stability from 100 to 500 zeros suggests the elevated ratio is persistent, not an artifact of the first few zeros.

|                             | 100 zeros | 500 zeros |
|-----------------------------|-----------|-----------|
| $\langle \tilde{r} \rangle$ | 0.641     | 0.646     |
| SE                          | 0.017     | 0.008     |

Table 3: Consistency check for  $\Delta$  zeros.

### 3.4 Statistical caveat

The deviation  $0.646 - 0.602 = 0.044$  represents 5.5 standard errors. If numerically accurate, this would be statistically significant. However:

1. No independent verification exists.
2. Finite-size bias for GUE is  $\approx +0.005$  at  $N = 500$ , much smaller than  $+0.044$ .
3. `lcalc` algorithmic parameters may introduce systematic bias.
4. No theoretical mechanism for “super-repulsion” is known.

We classify this as a **preliminary observation requiring independent verification**.

## 4 Discussion

### 4.1 Scope and limitations

This paper does **not**: present a new theorem; claim a new physical phenomenon; or provide a definitive answer about  $\Delta$  zero statistics.

It **does**: provide a fully reproducible computation; report a statistically significant discrepancy; and emphasize cross-validation.

### 4.2 Cross-validation recommendations

1. **PARI/GP**: `lfunzeros(L, 500)`.
2. **Mathematica**: `ZetaZero` for  $\zeta$ ; `LFunctionZero` for  $\Delta$ .
3. **Custom high-precision**: approximate functional equation with exact  $\tau(n)$  coefficients.

## 5 Conclusion

We have computed nearest-neighbor spacing ratios:  $\zeta$  zeros:  $0.615 \pm 0.012$  (consistent with GUE);  $\Delta$  zeros:  $0.646 \pm 0.008$  (above GUE, unverified). We invite cross-validation.

## References

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